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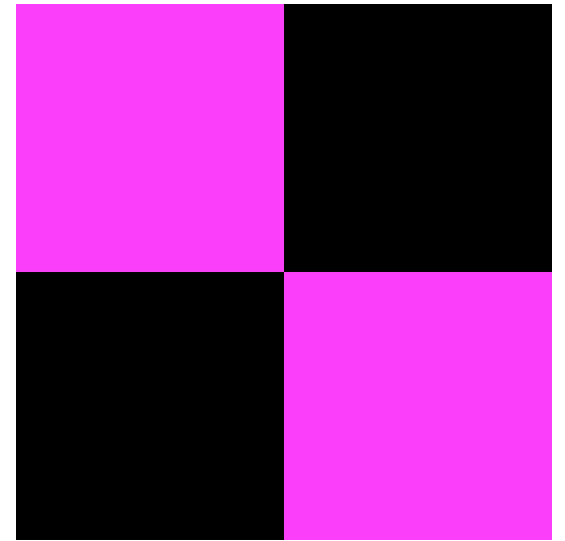


Module #: Title

<INSTRUCTOR NAME>

<WORKSHOP>

<DATES>



Instructor
affiliation
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Learning Objectives

- Understand the concept of supervised and unsupervised Machine Learning (ML).
- Understand what ML can and cannot do.
- Understand how to measure performance of ML models.
- Understand the concepts of overfitting and underfitting.



Supervised and Unsupervised Learning

- Essentially, every Machine Learning (ML) application can be tagged either as **Supervised Learning** or **Unsupervised Learning**.
- These are two fundamental concepts that are defined by how ML uses its data.



Supervised Learning

- Supervising learning is defined by the use of labelled data.
- You know beforehand which are the features **X** that describe a target **y**.
- Supervision comes from observing **y** during training.
- The algorithm learns the relationship between **X** and **y**.
- Then, it attempts to generalize the learned pattern to unseen **X**.



Supervised Learning

- Two main families of supervised learning.
- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)



Supervised Learning

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- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)

Problem: Predict whether a patient will survive 30 days



Supervised Learning

- Two main families of supervised learning.
- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)

Problem: Predict how many days a patient will survive



Supervised Learning

- Two main families of supervised learning.
- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)

Problem: Predict a patient's systolic blood pressure



Supervised Learning

- Two main families of supervised learning.
- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)

Problem: Predict cancer stage I, II, III, or IV.



Supervised Learning

- Two main families of supervised learning.
- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)

Problem: Predict pain-score (0-10)



Example: Supervised Learning, Classification

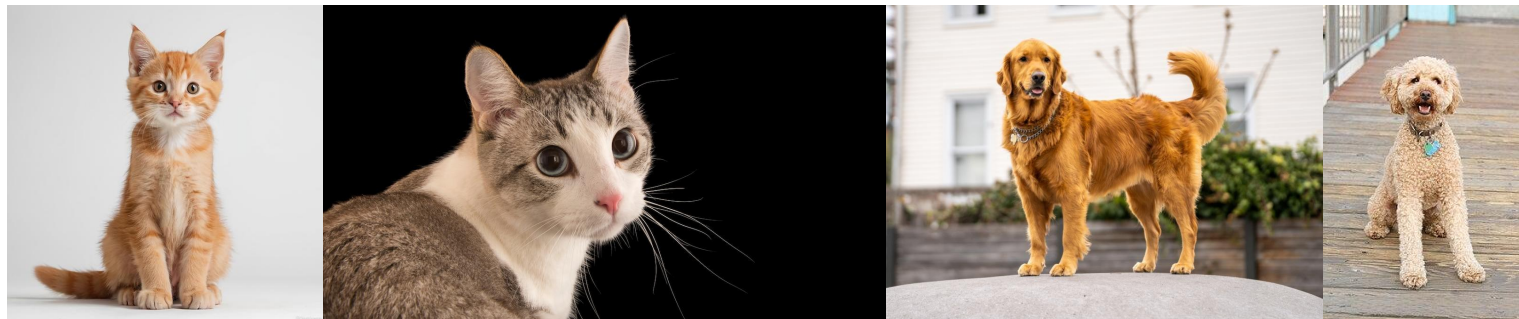
- **y** (binary; cat or dog; 1/0)
- **X** (features that describe y)

Cat

Dog

Ear shape

Training: We analyze n pictures of cats and dogs and try to find a ear shape that describes cats and dogs.





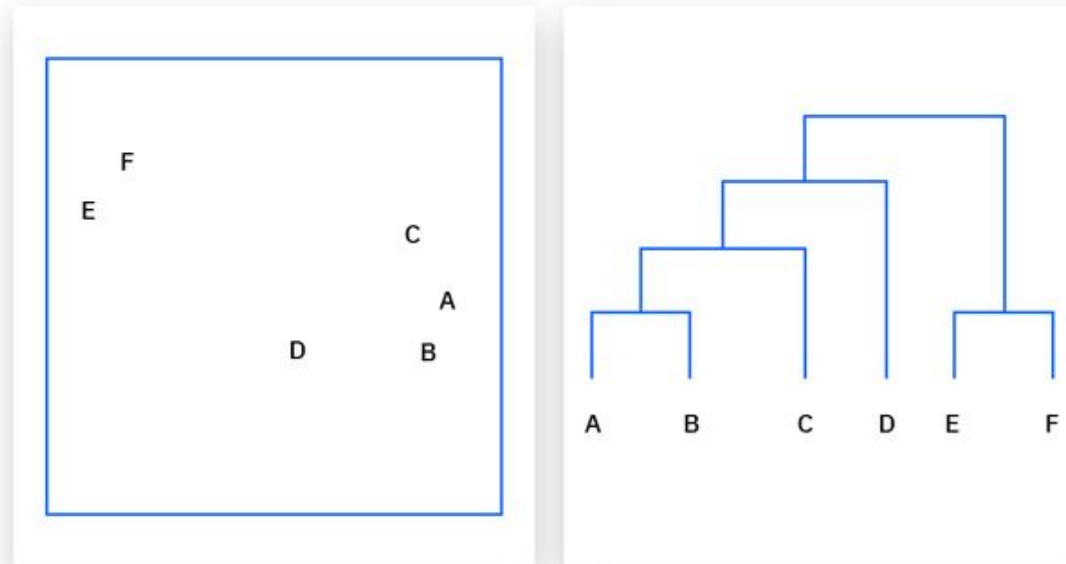
Unsupervised Learning

- Analyze unlabeled data sets to discover hidden patterns.
- There is no target label (**y**).
- Generally requires the involvement of someone familiar with the data.



Unsupervised Learning

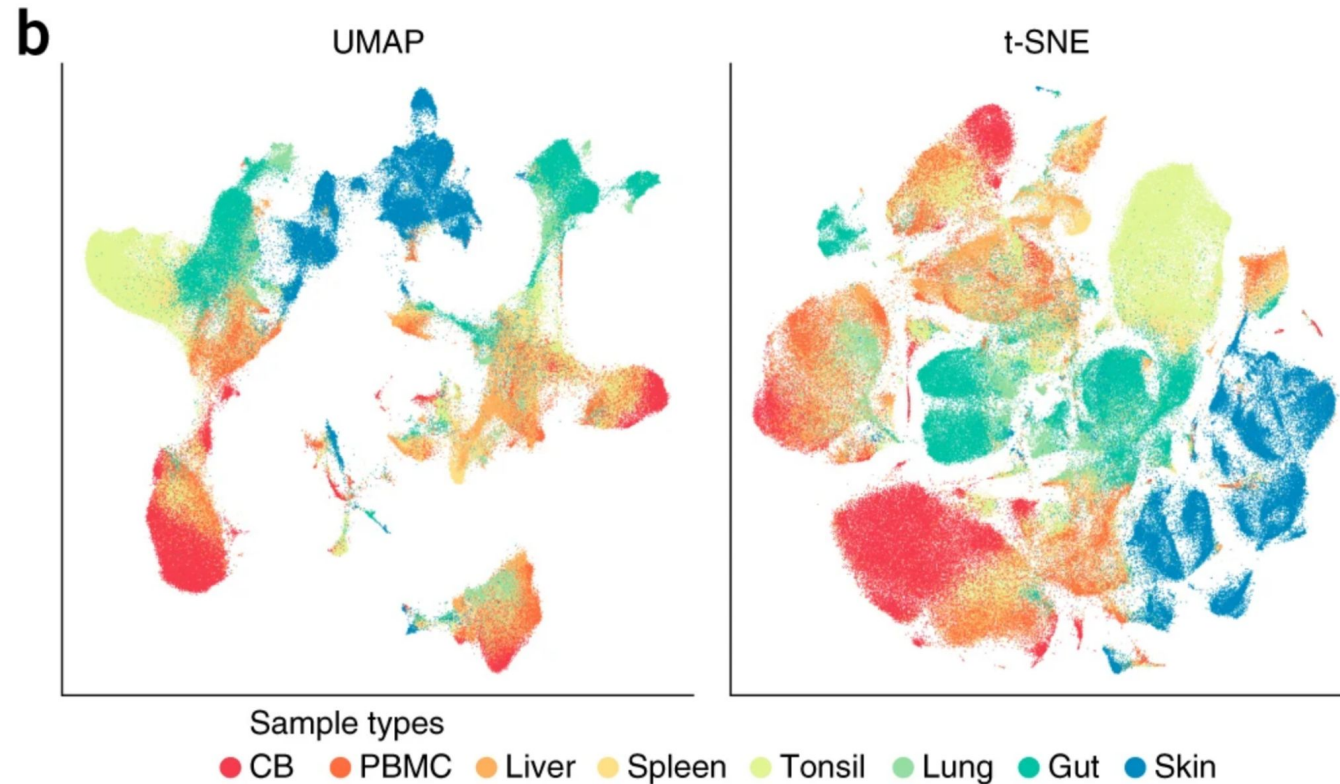
- Examples of algorithms:
 - **Clustering (day 3 with Mackenzie Tapp)**





Unsupervised Learning

- Examples of algorithms:
 - **Dimensionality reduction**





Measuring Performance (classification)

- Key Performance Indicators (KPI)
 - **Confusion Matrix**
 - Track correct predictions: True Positives (TP) and True Negatives (TN)
 - Track incorrect predictions: False Positives (FP) and False Negatives (FN)

Observed	Predicted	
	TP	FN
	FP	TN

Problem: *Prediction of having a disease*

TP: *Have the disease, was predicted as having*

FP: *Do not have the disease, was predicted as having*

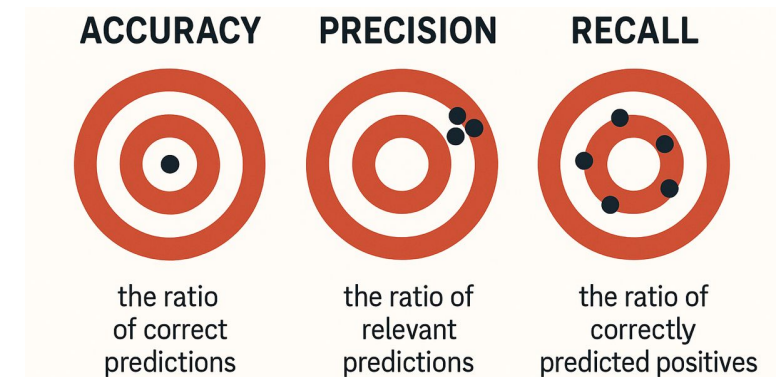
FN: *Have the disease, was predicted as not having*

TN: *Do not have the disease, was predicted as not having*



Measuring Performance (classification)

- Key Performance Indicators (KPI)
 - **Confusion Matrix**
 - **Accuracy** = Overall fraction of correct predictions out of all predictions.
 - $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$
 - **Precision** = Fraction of correct positive predictions
 - $\text{Precision} = \frac{TP}{TP+FN}$
 - **Recall (sensitivity)** = Fraction of actual positive cases the model captured.
 - $\text{Recall} = \frac{TP}{TP+FN}$
 - **Specificity** = Fraction of actual negative cases the model captured.
 - $\text{Specificity} = \frac{TN}{TP+FN}$





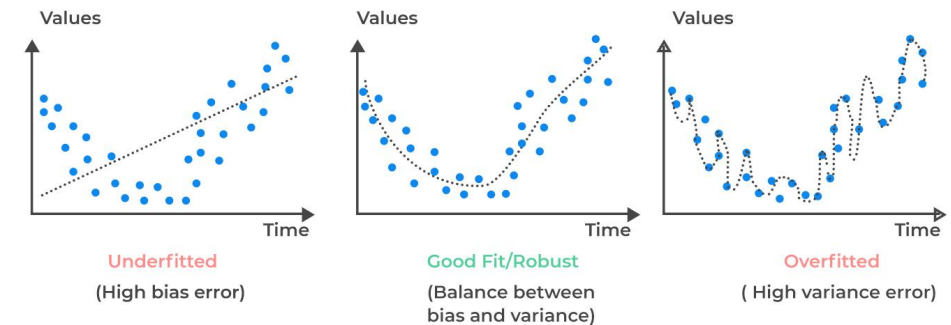
Measuring Performance (Regression)

- We measure the difference between the ground truth targets (actual) and the model's predicted values (predicted).
- These differences are referred to as residuals.



Over/Underfitting

- Scenario: Student training for an exam.
 - **Underfitting (the unprepared student)**
 - The student misses fundamental principles. Scores poorly on practice problems (high training error) and fail the final exam (high test error).
 - **Overfitting (the rot-memorizer)**
 - The student memorizes every question and answer from the practice exam. Scores 100% on practice problems (low training error), but fails when given slightly modified questions (high test error).





Coffee Break & Networking Session

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